Different Information Worlds

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An organization’s information usually exists in **two very different worlds**:

1. **Operational Systems (OLTP – Online Transaction Processing)**
2. **Data Warehouse (OLAP – Online Analytical Processing)**

They may store the same business data, but they serve completely different purposes.

Operational Systems – Where Data Is Created
Operational systems are the systems that run the daily business.
Examples:

- Taking orders
- Registering customers
- Logging complaints
- Processing payments

These systems are built for:

- Speed
- Accuracy
- Handling one transaction at a time
- Supporting repetitive tasks

Users interact with **one record at a time**.
Think of a cashier scanning one product.
Or a support agent updating one complaint.
The goal here is **execution**, not analysis.
Technically:

- Highly normalized databases (often 3NF)
- Optimized for inserts, updates, deletes
- Short, fast transactions

This is the “engine room” of the company.

Data Warehouse – Where Data Is Analyzed
The data warehouse serves a completely different audience.
Its users:

- Analysts
- Executives
- Managers
- Data scientists

They don’t care about one transaction.
They ask questions like:

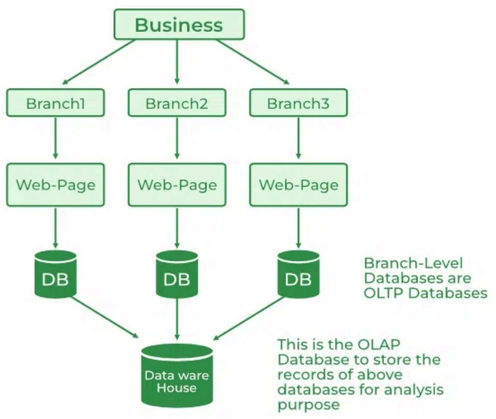
- How many orders were placed this week vs last week?
- Why did customer churn increase?
- Which product category is growing fastest?

These questions require:

- Scanning thousands or millions of rows
- Aggregating data
- Comparing across time
- Summarizing patterns

Users rarely retrieve a single row.

They retrieve **patterns across many rows**.
And their questions constantly change.
That’s a key point:
Operational tasks are repetitive.
Analytical questions are unpredictable.
The goal here is **understanding**, not execution.



https://www.geeksforgeeks.org/dbms/difference-between-olap-and-oltp-in-dbms/

OLAP (Online Analytical Processing) and OLTP (Online Transaction Processing) are both integral parts of data management, but they have different functionalities.

- OLTP focuses on handling large numbers of transactional operations in real time, ensuring data consistency and reliability for daily business operations.
- OLAP is designed for complex queries and data analysis, enabling businesses to derive insights from vast datasets through multidimensional analysis.

- **OLTP DBs** → normalize → good for transactions
- **OLAP / DWH** → denormalize → good for analysis and reporting

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Even if both systems use the same business data, they differ in:

- User type
- Query behavior
- Performance requirements
- Data structure
- Frequency of change

Operational systems:

- Focus on current state
- Optimize for writes
- Handle small, fast transactions

Data warehouses:

- Focus on history
- Optimize for reads
- Handle large, complex queries

This is not just a hardware separation issue.

Moving an operational database to another server does not automatically transform it into a data warehouse.

- OLTP = “doing the work” (write)
- OLAP = “analyzing the work” (read)

If the structure remains transaction-oriented:

- Queries become complex
- Joins multiply
- Performance degrades
- Business users struggle

True analytical environments must be designed specifically for analytical needs.
